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In [7]: import pandas as pd
df=pd.read_csv('Housing.csv')
df.head()
```

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Out[7]:
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	price	area	bedrooms	bathrooms	stories	mainroad	guestroom	basement	hotwater
0	13300000	7420	4	2	3	yes	no	no	
1	12250000	8960	4	4	4	yes	no	no	
2	12250000	9960	3	2	2	yes	no	yes	
3	12215000	7500	4	2	2	yes	no	yes	
4	11410000	7420	4	1	2	yes	yes	yes	

```
In [8]: num_cols=['area','bedrooms','bathrooms','stories','parking']
outliers={}
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In [9]: # Finding Outliers using IQR method
for col in num_cols:
    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outlier_rows = df[(df[col] < lower_bound) | (df[col] > upper_bound)]
    outliers[col] = outlier_rows.shape[0]
    print(f"{col}: {outlier_rows.shape[0]} outliers found")
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area: 12 outliers found
bedrooms: 12 outliers found
bathrooms: 1 outliers found
stories: 41 outliers found
parking: 12 outliers found
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In [10]: # Outliers using Z-scores
import numpy as np
from scipy.stats import zscore
z_scores = np.abs(df[num_cols].apply(zscore))
outliers_z = (z_scores > 3)
for col in num_cols:
    print(f"{col}: {outliers_z[col].sum()} outliers found")
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area: 7 outliers found
bedrooms: 2 outliers found
bathrooms: 11 outliers found
stories: 0 outliers found
parking: 0 outliers found
```

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In [11]: # Outliers using Mod_Z-Score
def modified_z_score(series):
    median = np.median(series)
    mad = np.median(np.abs(series - median)) # Median Absolute Deviation
    modified_z = 0.6745 * (series - median) / mad
    return np.abs(modified_z)

modified_z_scores = df[num_cols].apply(modified_z_score)
outliers_modified = (modified_z_scores > 3.5)
for col in num_cols:
    print(f"{col}: {outliers_modified[col].sum()} outliers found")

area: 7 outliers found
bedrooms: 245 outliers found
bathrooms: 144 outliers found
stories: 0 outliers found
parking: 246 outliers found
```

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In [ ]:
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